

CS736 - Assignment-2

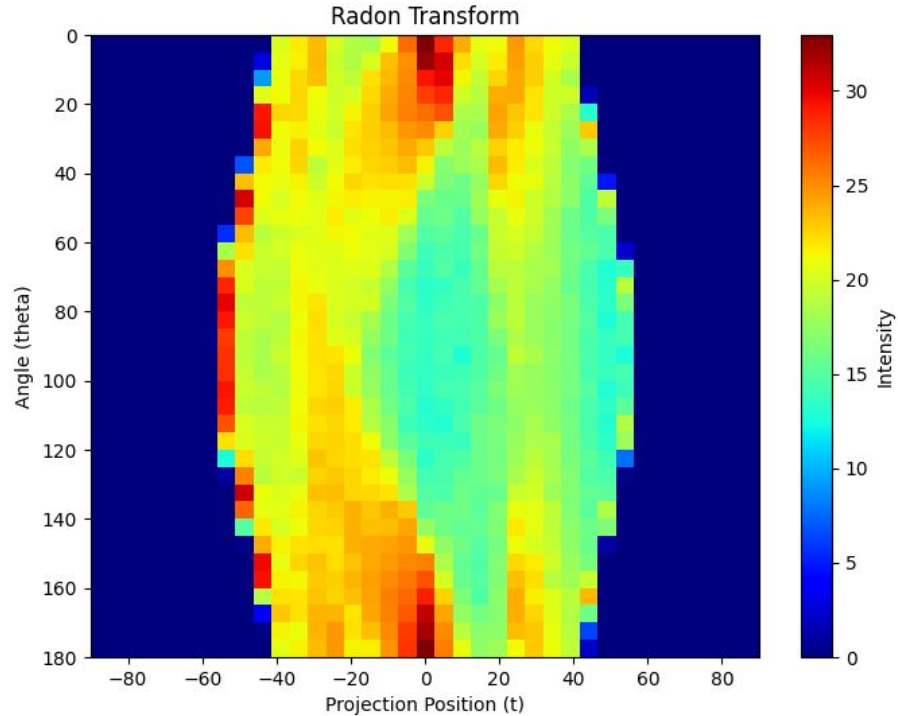
Team Members

1. Atharv Shingewar (24M0834)
2. Russel Abreo (24M2107)

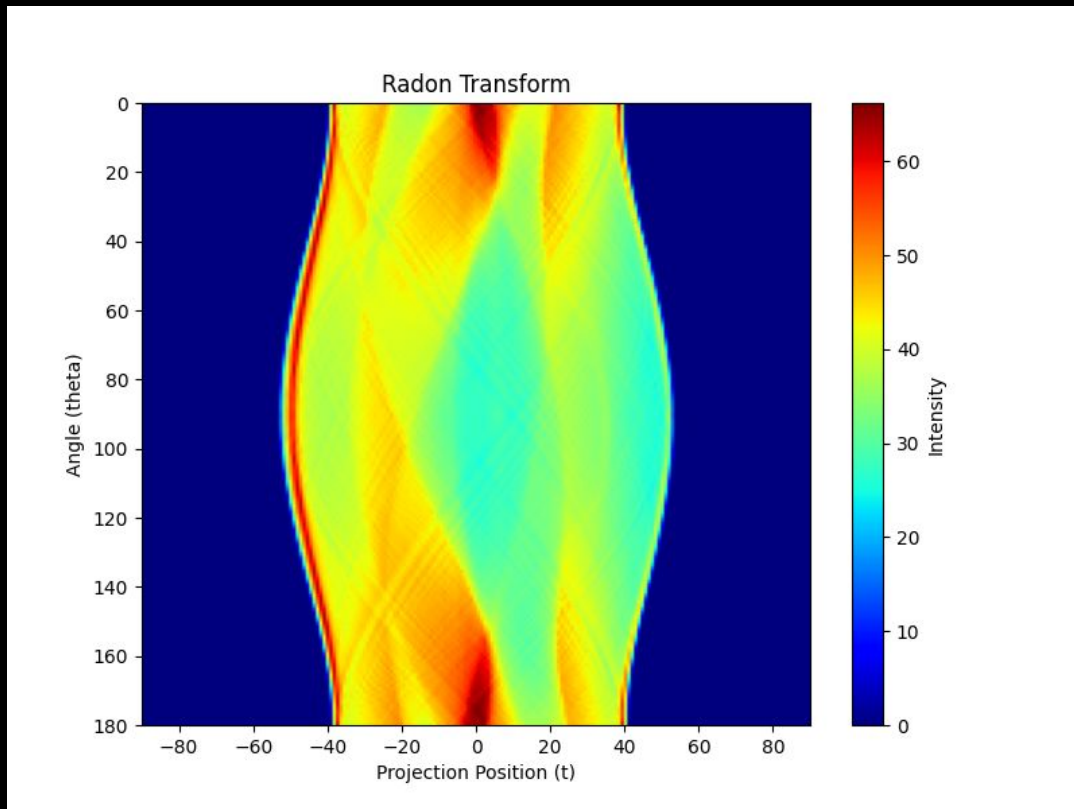
Under the Guidance of :

Prof. Suyash Awate

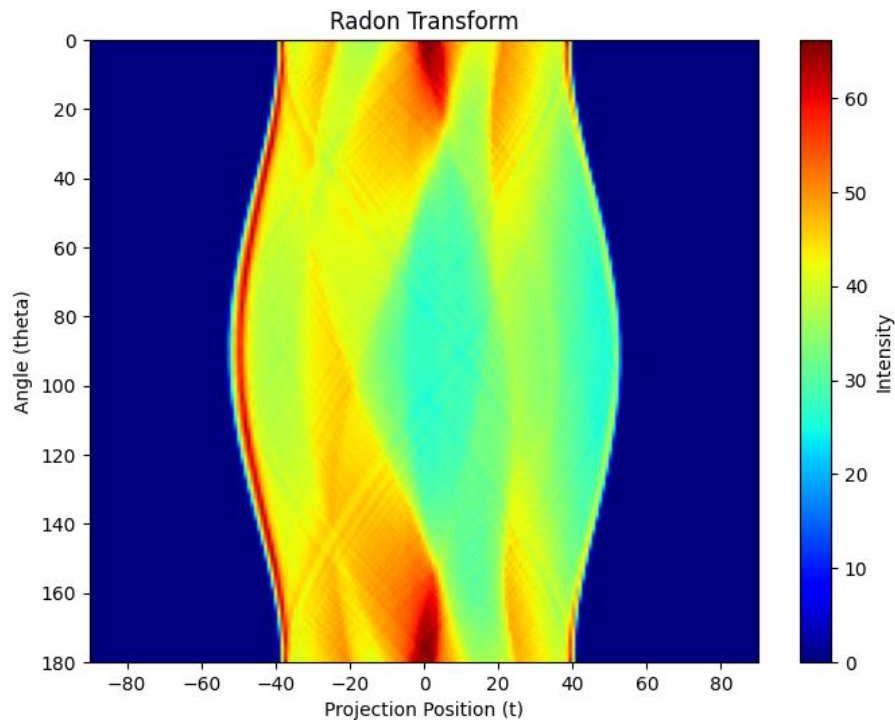
Q1b - Radon Transform



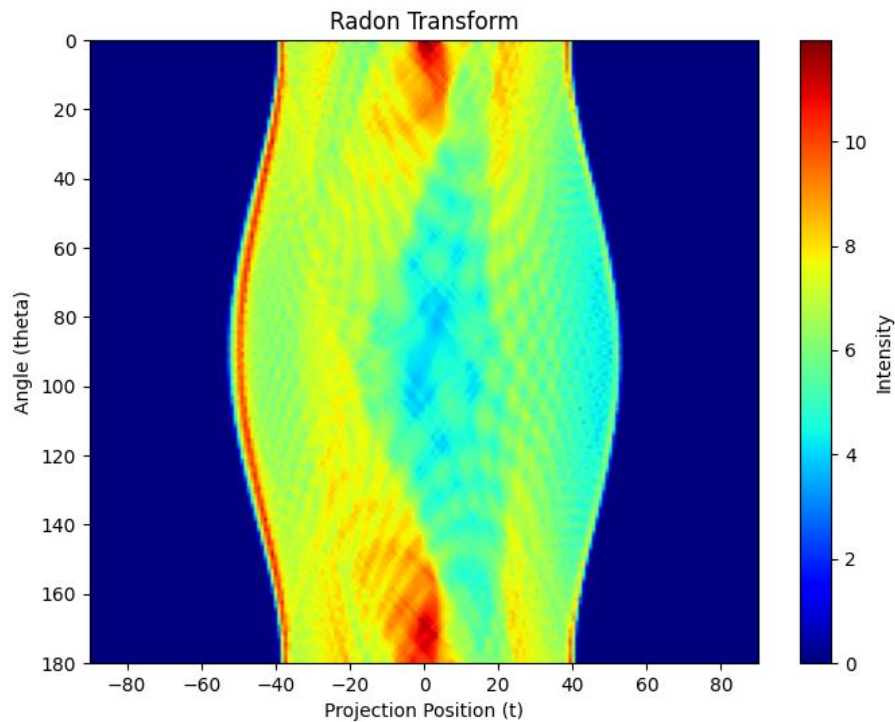
Q1c - Radon Transforms | $s = 0.5$



Q1 - Radon Transforms | $s = 1$

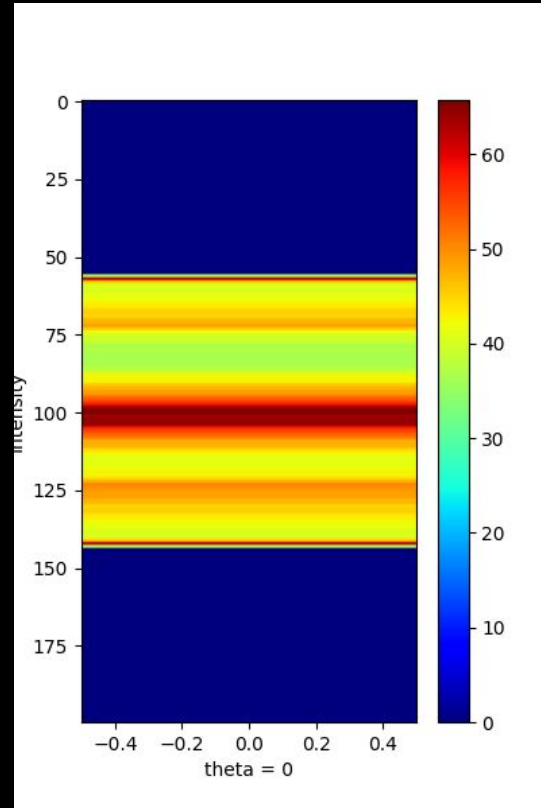


Q1 - Radon Transforms | $s=3$



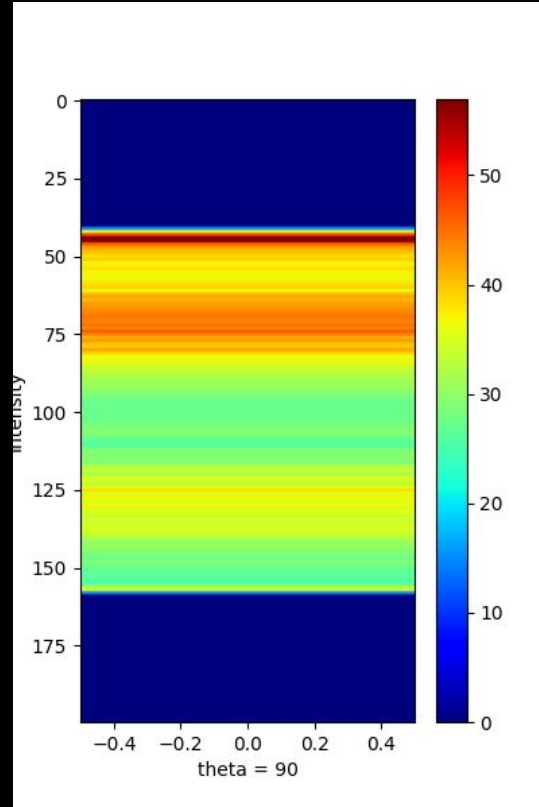
Q1 - 1D Plot Radon Transforms

$s=0.5$, $\theta=0$



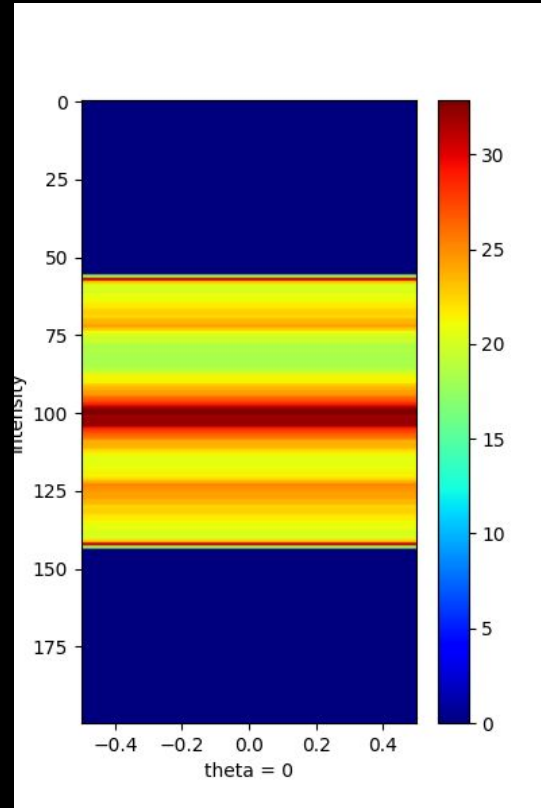
Q1 - 1D Plot Radon Transforms

$s=0.5$, $\theta=90$



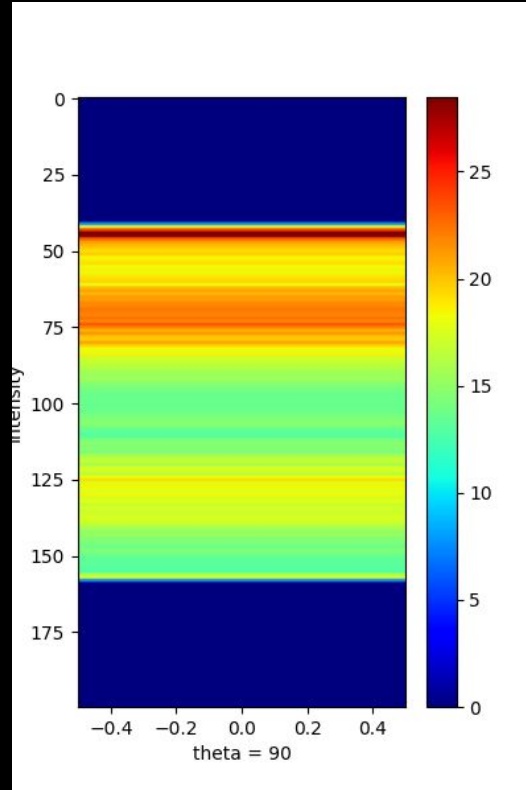
Q1 - 1D Plot Radon Transforms

$s=1$, $\theta=0$



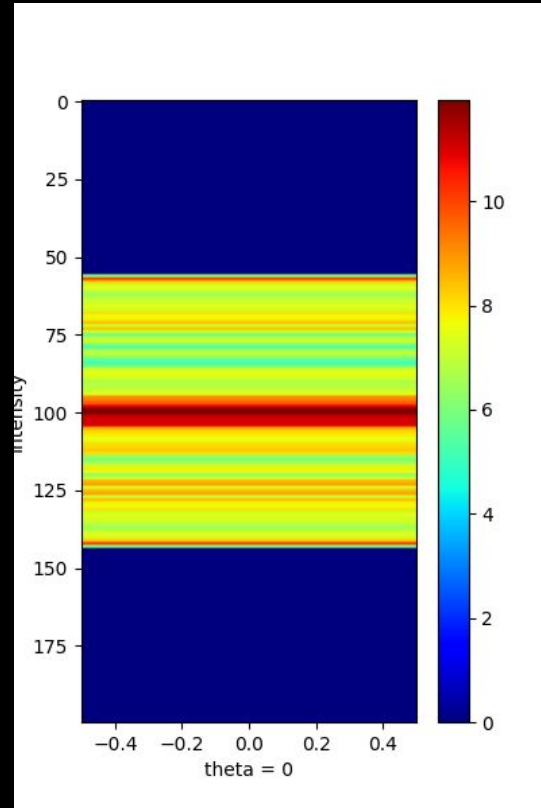
Q1 - 1D Plot Radon Transforms

$s=1$, $\theta=90$



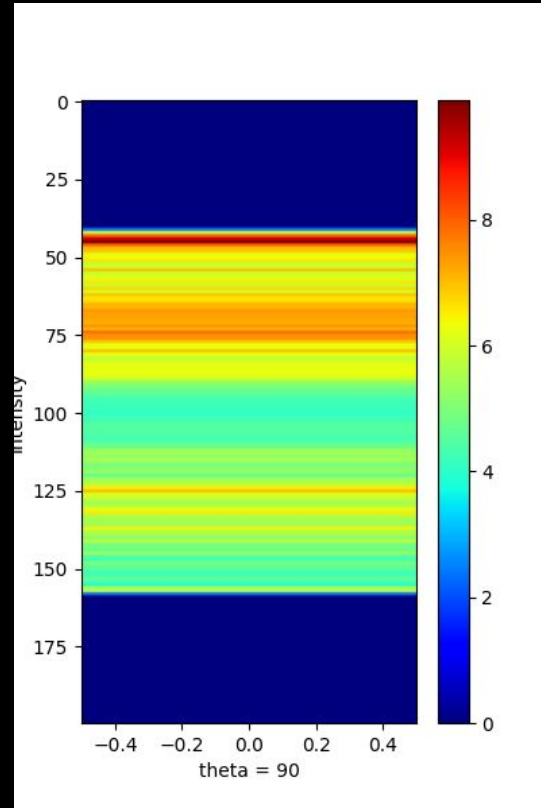
Q1 - 1D Plot Radon Transforms

$s=3$, $\theta=0$



Q1 - 1D Plot Radon Transforms

$s=3$, $\theta=90$



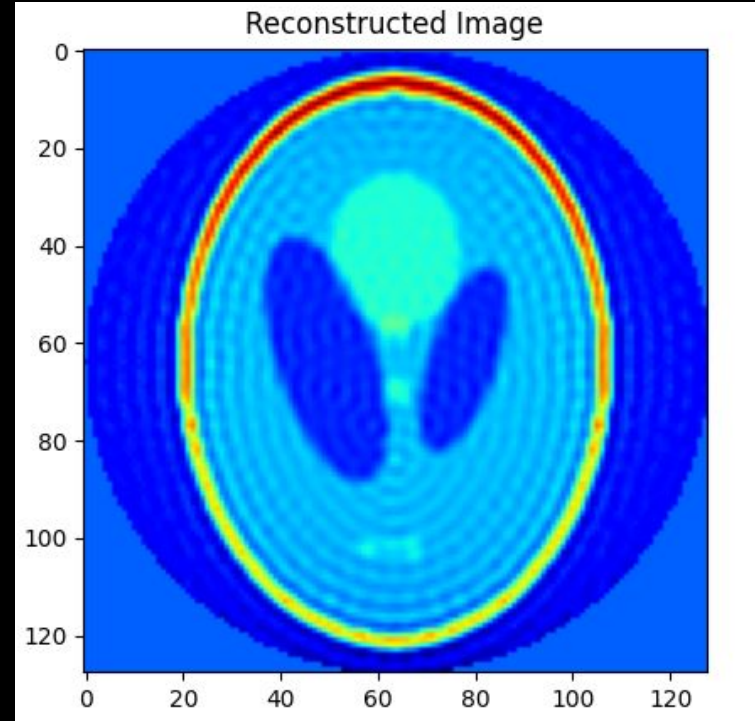
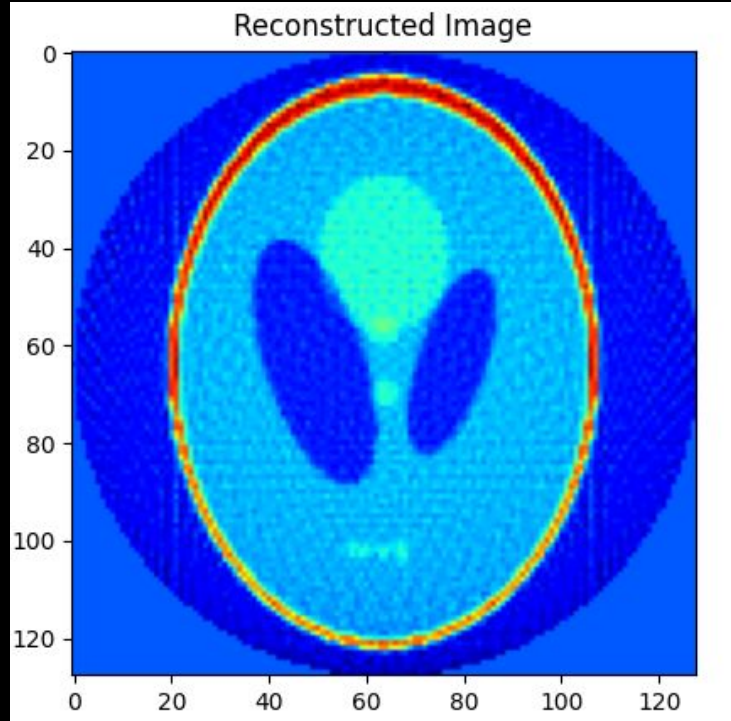
Q1 - d

- **Too large delta t:** May undersample the projection data, which can lead to missing finer details while reconstruction of images.
- **Too small delta t:** Can help with catching finer details but with extremely small delta_t may cause increase in noise.
- **Too large delta_theta:** With very large delta_theta, certain angular components may be missed during reconstruction of images.
- **Too small delta_theta:** Will cause increase in scan times, reconstruction will be expensive(compute) and this will also lead to high doses of radiation.

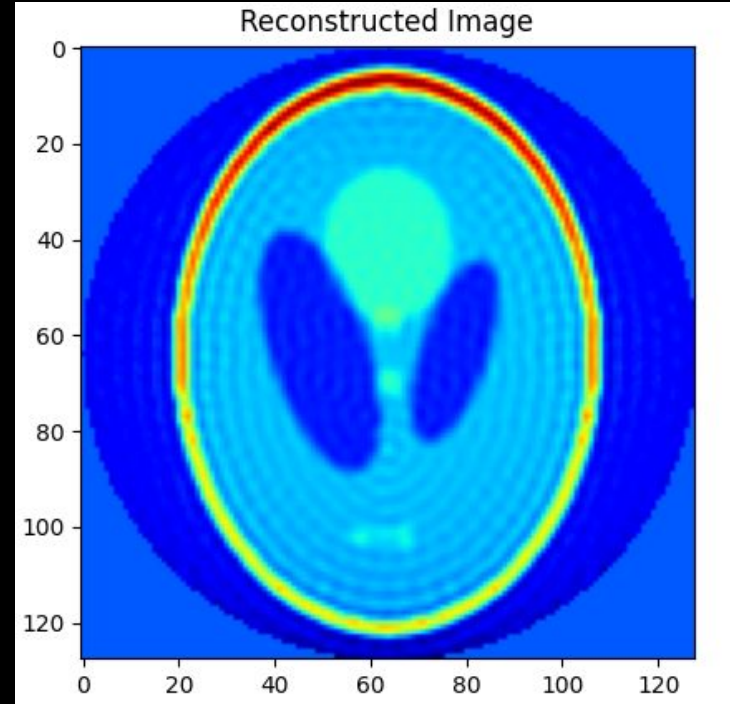
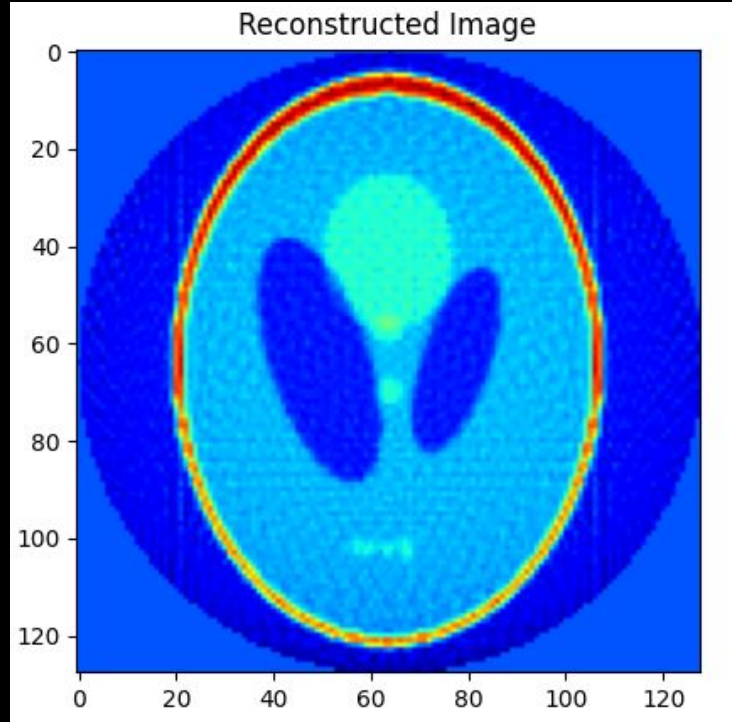
Q1 - e

- Smaller pixel size would yield better detail at the cost of increased computation.
- For $\Delta s \gg 1$ pixel width : There would be few sampled points due to the large step size. It would lead to loss of fine details during the reconstruction process . This would be a good choice when computation is restriction.
- For $\Delta s \ll 1$: With more sample points there would be an improved reconstruction. But it would increase memory usage and computational cost.

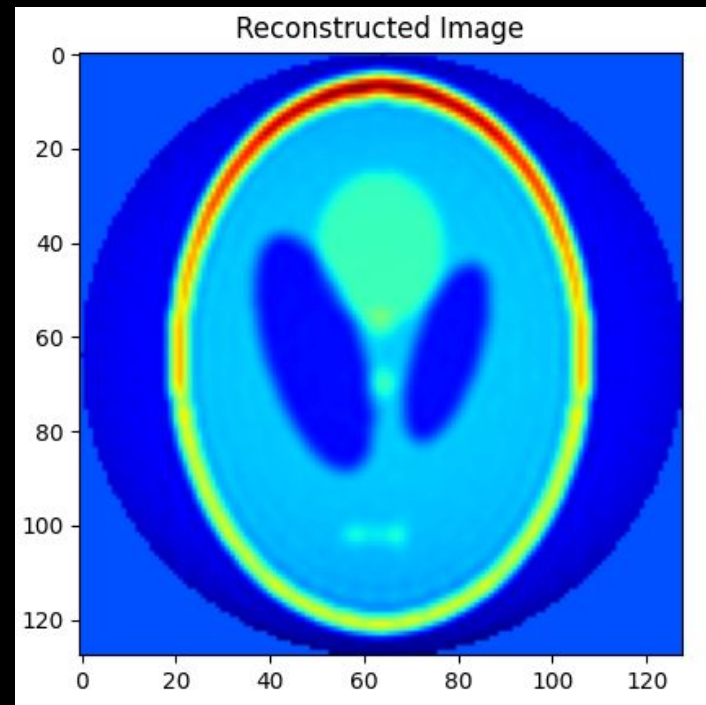
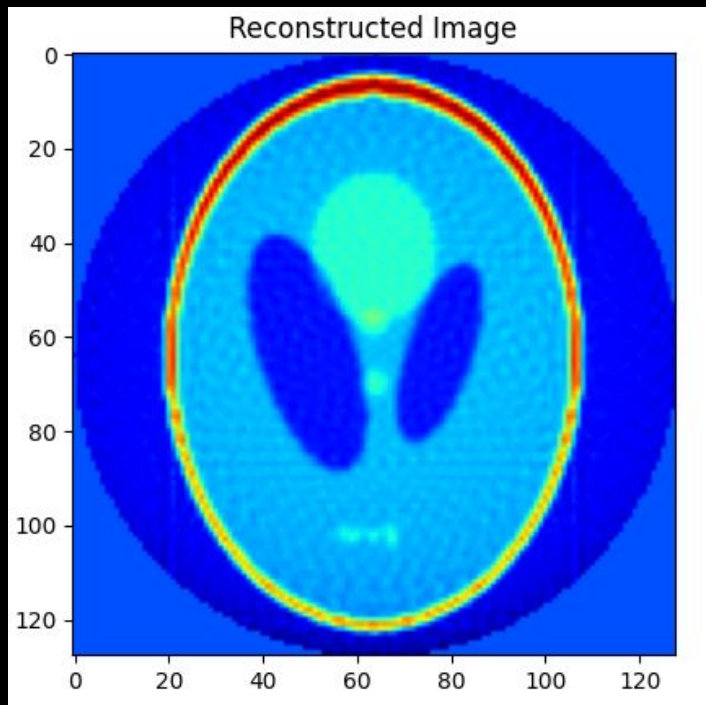
Q2a. Ram-Lak ($L = W_{\max}$, $L = W_{\max} / 2$)



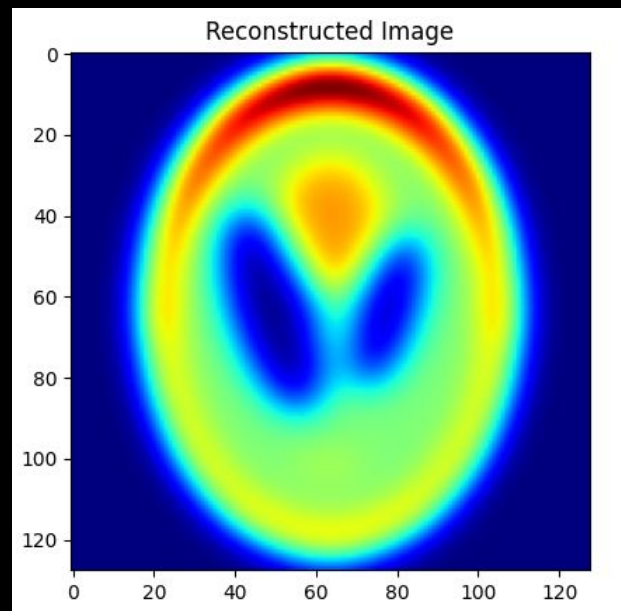
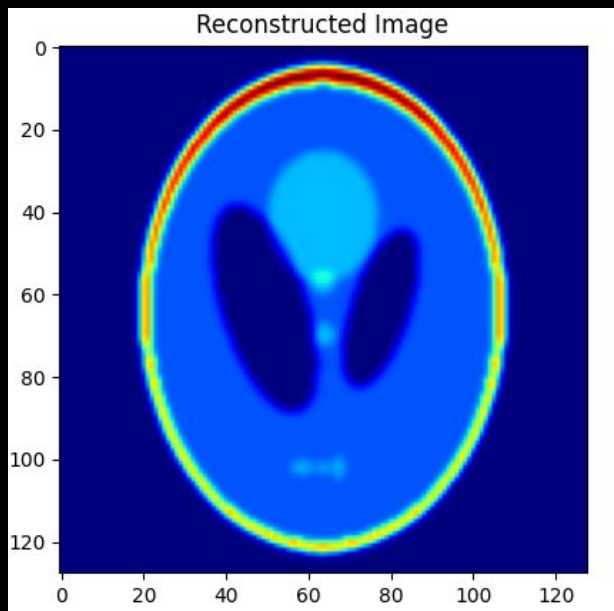
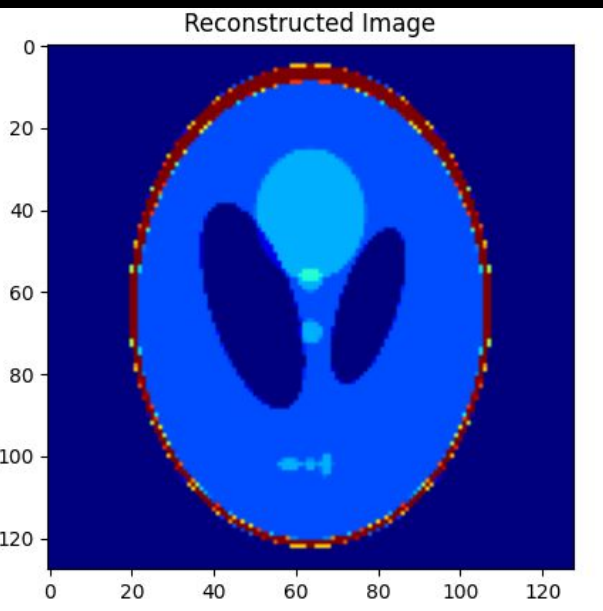
Q2a. Shepp-Logan (L = Wmax, L = Wmax / 2)



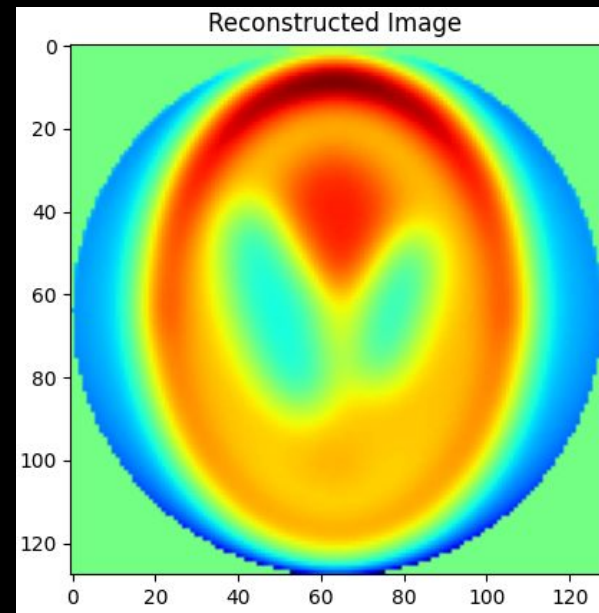
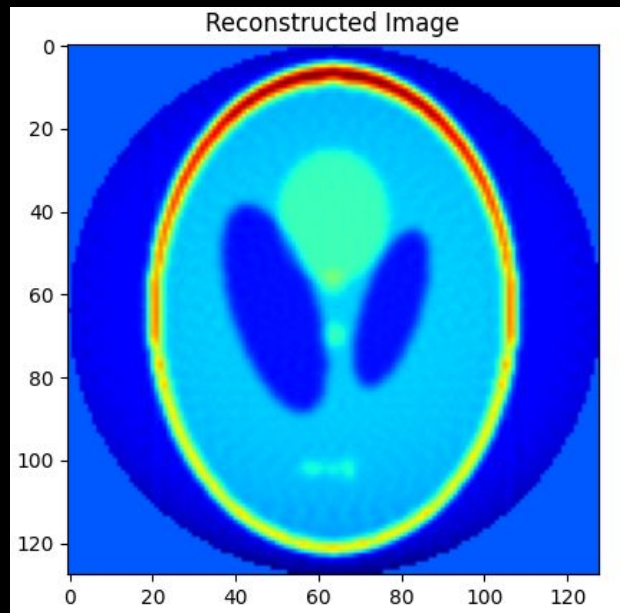
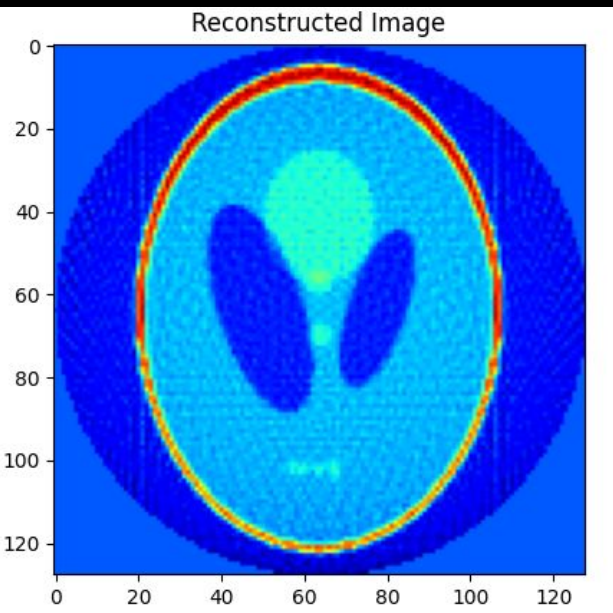
Q2a. Cosine ($L = W_{\max}$, $L = W_{\max} / 2$)



Q2b. Blurred Images(S0[Unblurred], S1, S5)



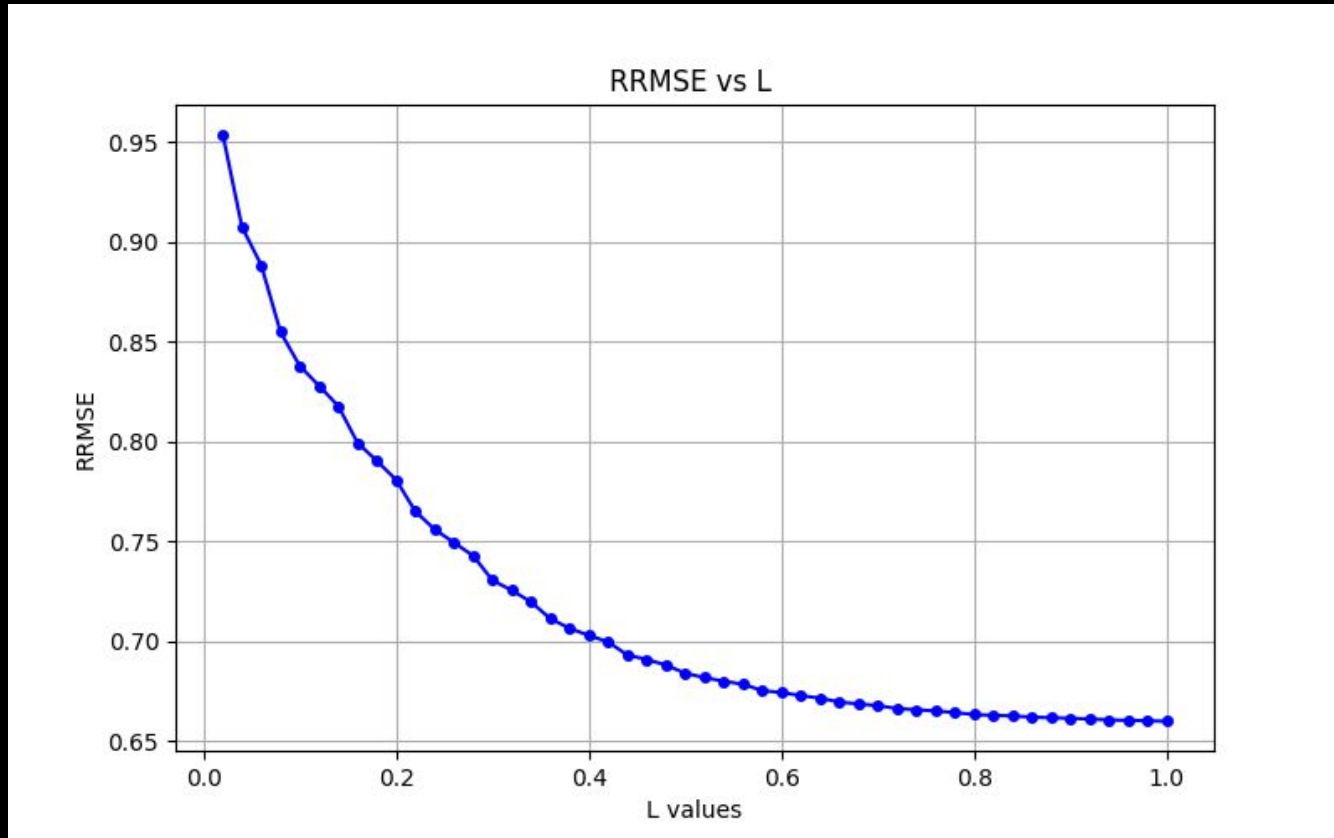
Q2b.Reconstructed Blurred Images(R0, R1, R5)



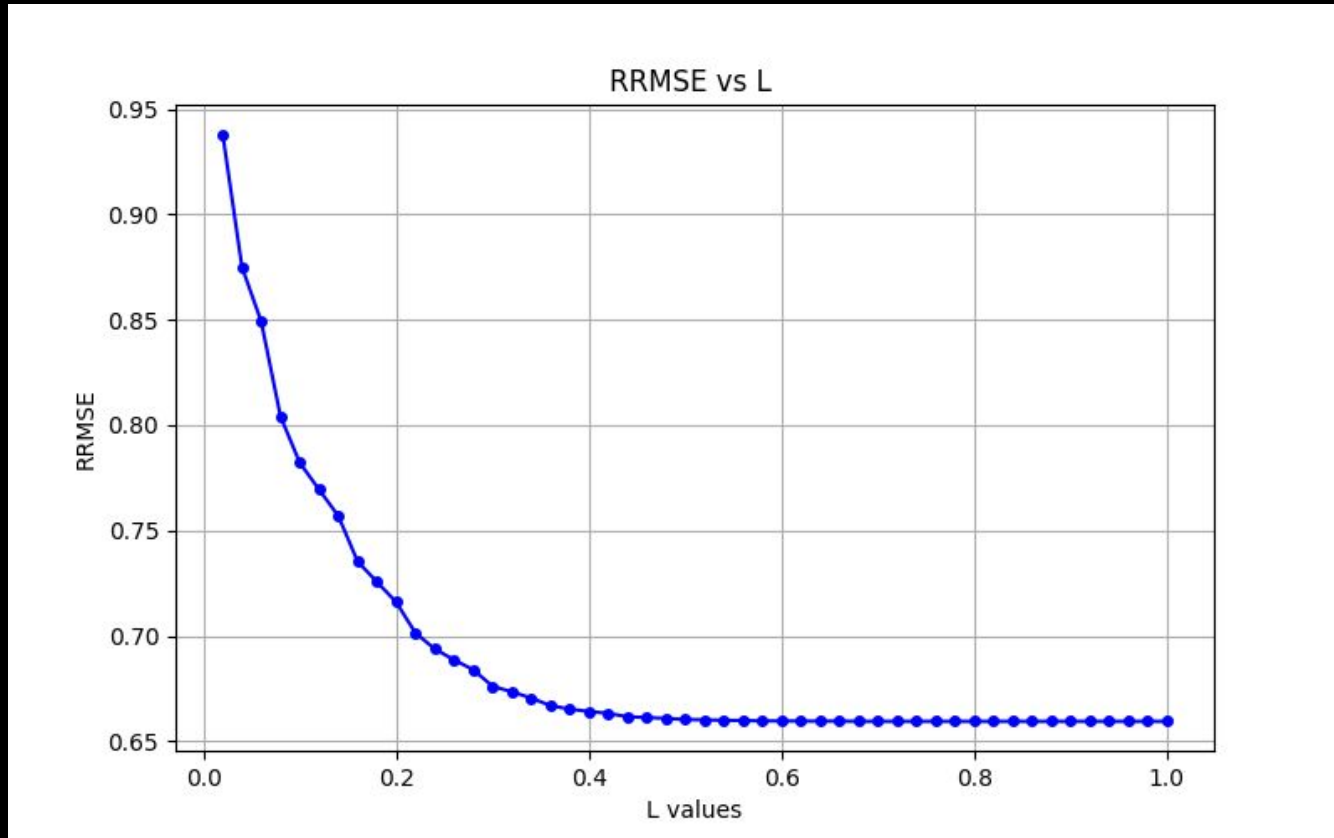
Q2b. RRMSE Values(R0, R1, R5)

1. RRMSE for phantom : 0.6598
2. RRMSE for blurred_1 : 0.6594
3. RRMSE for blurred_5 : 0.7382

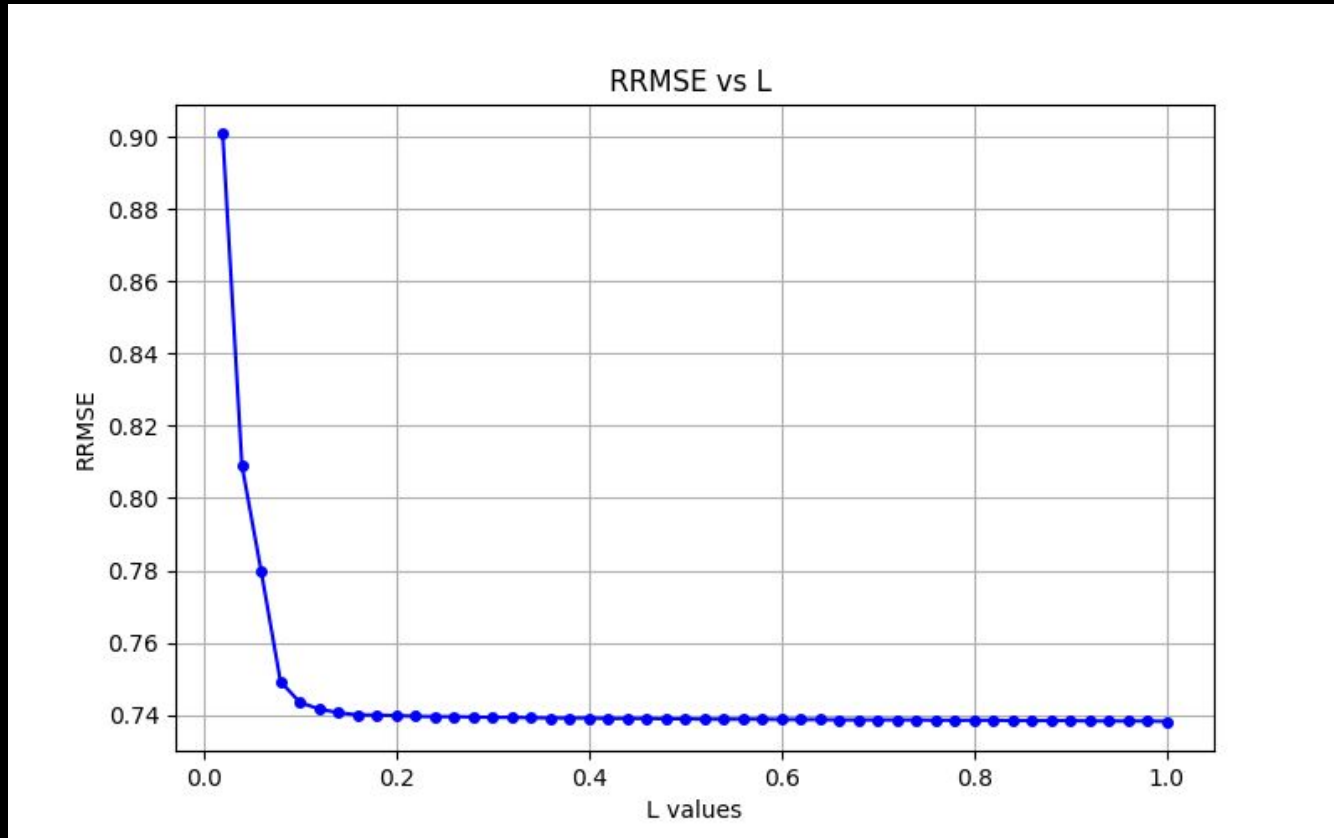
Q2c. RRMSE values as a function of L: R0



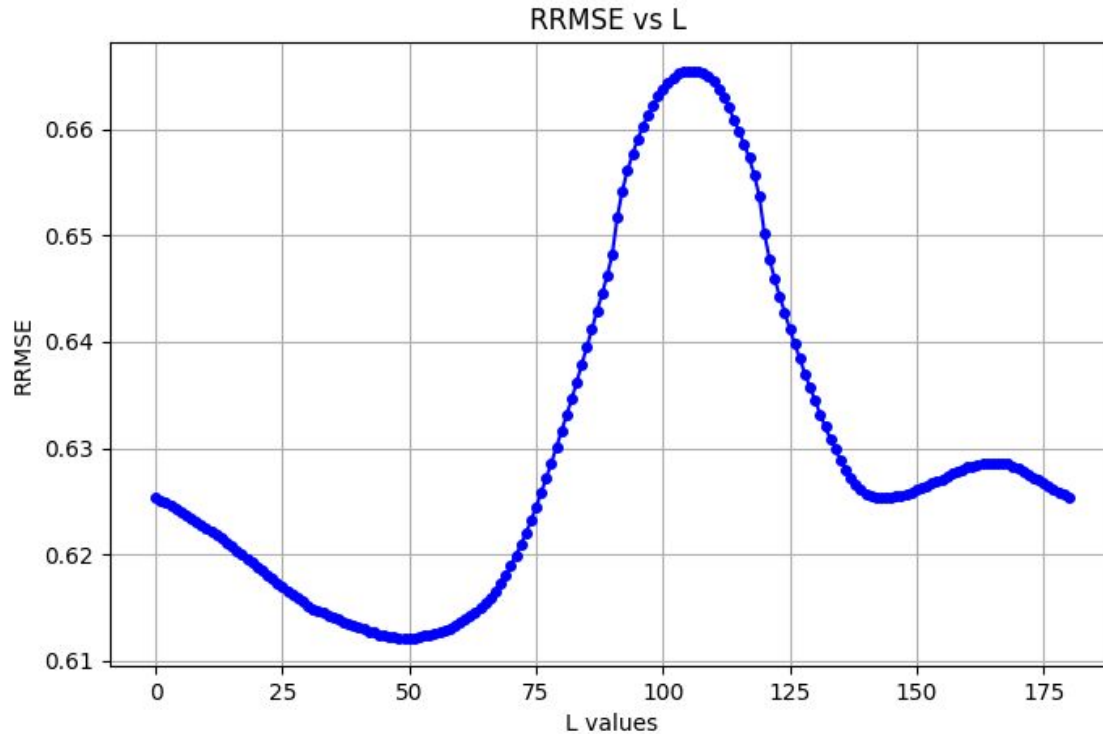
Q2c. RRMSE values as a function of L: R1



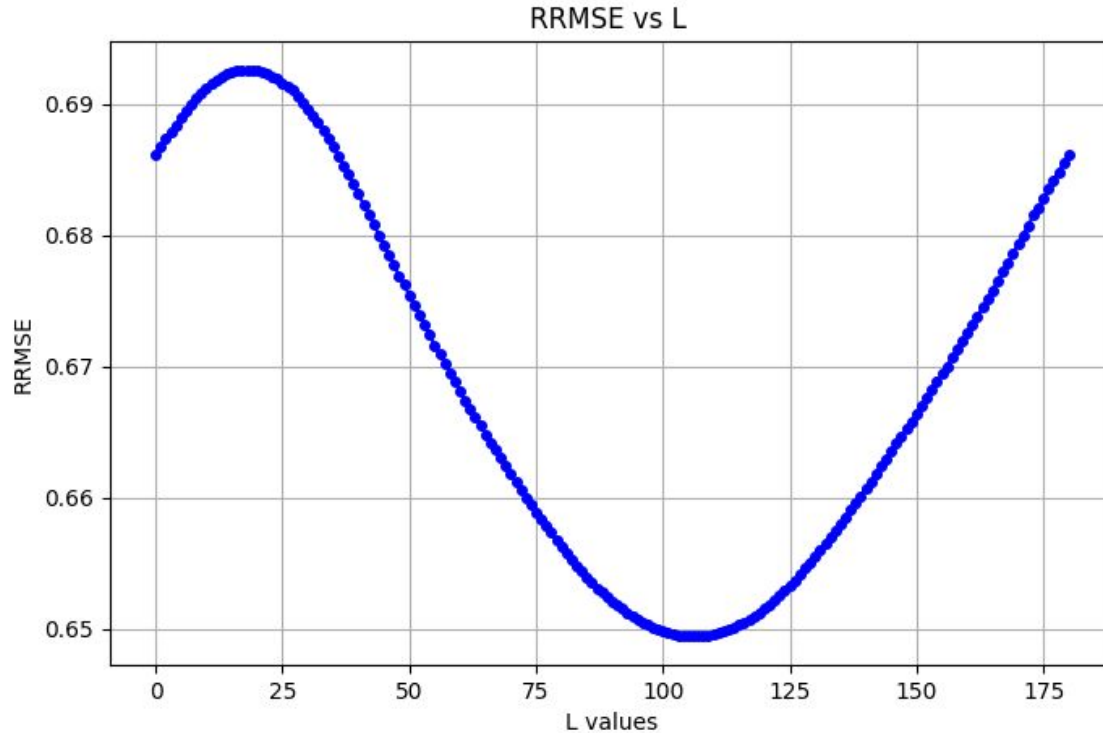
Q2c. RRMSE values as a function of L: R5



Q3a - RRMSE chest CT

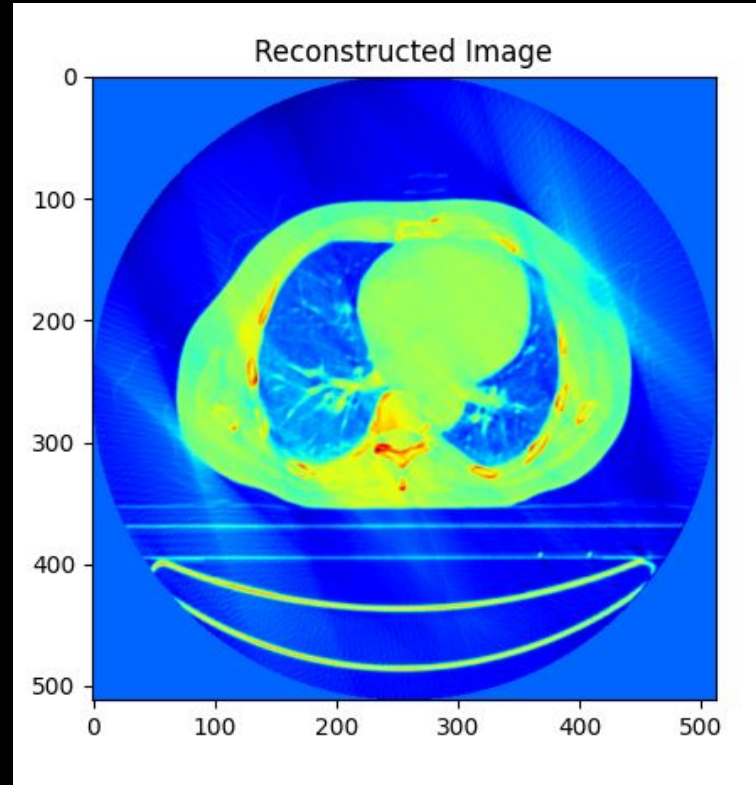


Q3a - RRMSE Phantom



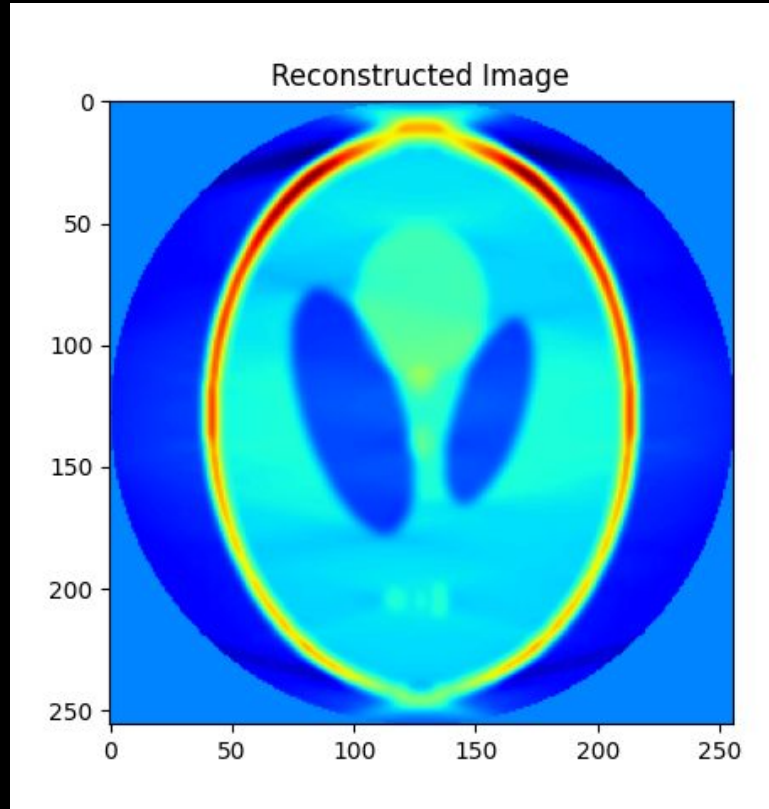
Q3b - Reconstructed chest CT

theta = 48, RRMSE = 0.6122

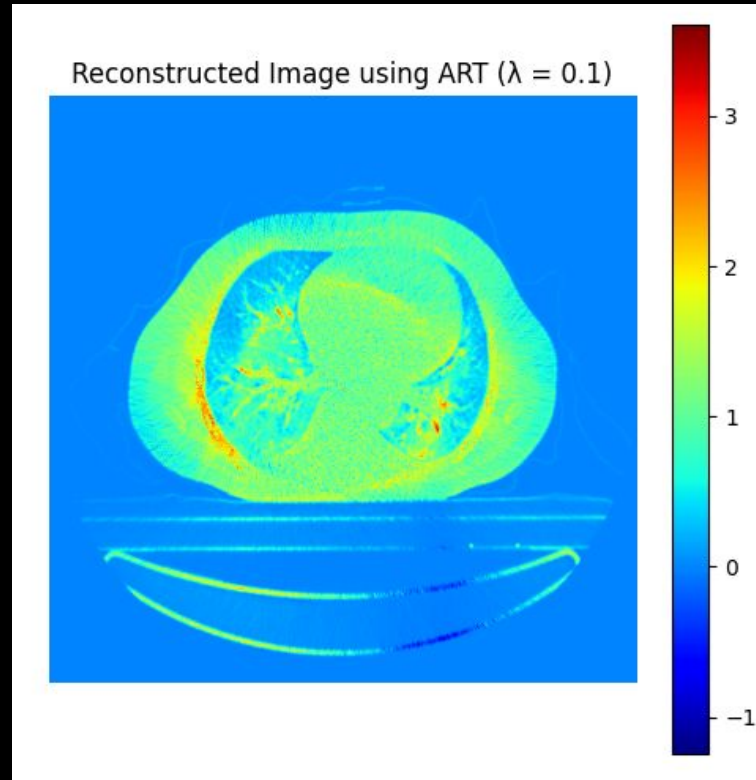


Q3b - Reconstructed Phantom

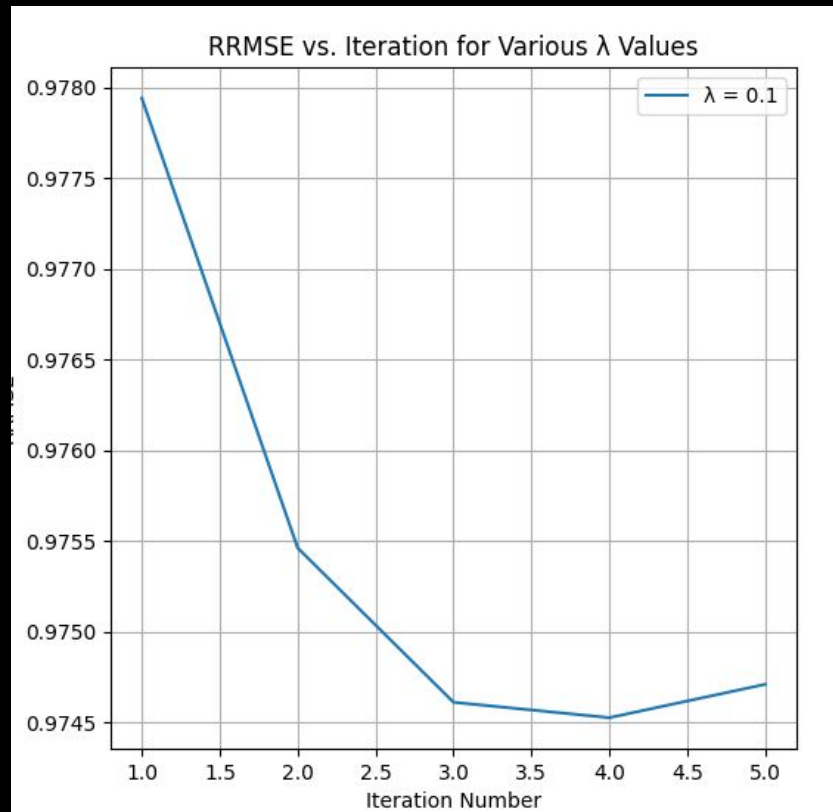
theta = 105, RRMSE = 0.6494



Q4 Reconstructed Image



Q4: RRMSE vs Iterations



Thank You